

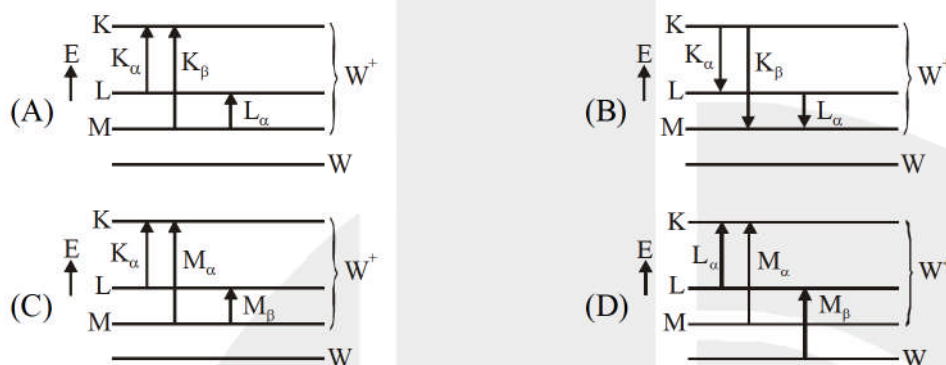
**[SINGLE CORRECT CHOICE TYPE]**

- Q.1 The binding energy of a heavy nucleus is about 7 million electron volts per nucleon, whereas the binding energy of a medium-weight nucleus is about 8 million electron volts per nucleon. Therefore, the total kinetic energy liberated, when a heavy nucleus undergoes fission with identical daughter nuclei, is most nearly  
(A) 1876 MeV (B) 938 MeV (C) 200 MeV (D) 8 MeV
- Q.2 When 1 cm thick surface is illuminated with light of wavelength  $\lambda$ , the stopping potential is  $v$ . When the same surface is illuminated by light of wavelength  $2\lambda$ , the stopping potential is  $v/3$ . Threshold wavelength for metallic surface is  
(A)  $4\lambda/3$  (B)  $4\lambda$  (C)  $8\lambda/3$  (D)  $6\lambda$
- Q.3 A nucleus of nuclear density  $\rho$  disintegrates into two daughter nuclei with masses in the ratio 8 : 27. Density of the smaller nucleus is  
(A)  $\frac{2}{3}\rho$  (B)  $\frac{2}{5}\rho$  (C)  $\frac{8}{27}\rho$  (D)  $\rho$
- Q.4 The electron in a H - atom makes a transition from  $n_2 \rightarrow n_1$  where  $n_1$  and  $n_2$  are the principal quantum number of the states. Assume Bohr's model to be valid. The time period of the electron in initial state is 27 times than that in final state. Find the possible value of  $n_1$  and  $n_2$ , if only just first 10 states are in the observable range of instrument.  
(A) 1-2, 2-4, 3-6, 4-8, 5-10 (B) 1-3, 2-6, 3-9  
(C) 14, 28 (D) none
- Q.5 A small mirror of mass  $m$  is suspended by a light thread of length  $\ell$ . The angle through which the thread will be deflected when a short pulse of laser of energy  $E$  falls normally on the mirror is  
(A)  $\frac{2E}{mc\sqrt{2g\ell}}$  (B)  $\frac{mc\sqrt{g\ell}}{2E}$  (C)  $\frac{mc\sqrt{2g\ell}}{2E}$  (D)  $\frac{2E}{mc\sqrt{g\ell}}$
- Q.6 In the nuclear reaction  ${}_1\text{H}^2 + {}_1\text{H}^2 \rightarrow {}_2\text{He}^3 + {}_0\text{n}^1$ , Mass of deuterium atom = 2.014741 amu, Mass of  ${}_2\text{He}^3$  atom = 3.016977 amu, and Mass of neutron = 1.008987 amu. The Q-value of the reaction will be  
(A) 0.00352 MeV (B) 3.27 MeV (C) 0.82 MeV (D) 2.45 MeV
- Q.7 The  $k_\alpha$  wavelength of an element with atomic number  $Z$  is  $\lambda_1$ . The  $k_\alpha$  wavelength of an element with atomic number  $2Z$  is  $\lambda_2$ . How are  $\lambda_1$  and  $\lambda_2$  related?  
(A)  $\lambda_1 > 4\lambda_2$  (B)  $\lambda_2 = 4\lambda_1$  (C)  $\lambda_1 < 4\lambda_2$   
(D) Depending on  $z$ ,  $\lambda_1$  could be greater than or less than  $4\lambda_2$
- Q.8 What is the minimum kinetic energy needed by a hydrogen atom in ground state to produce a photon after collision with another identical hydrogen atom at rest?  
(A) 6.8eV (B) 3.4eV (C) 20.4eV (D) 2.04eV

Q.9 If the potential energy between electron and proton at a distance  $r$  is given by  $-\frac{ke^2}{3r^3}$ . Then by Bohr's theory, the velocity of electron in  $n^{\text{th}}$  orbit of hydrogen atom is

- (A)  $\frac{n^3 h^3}{8\pi^3 k m^2 e^2}$  (B)  $\frac{n^2 h^3}{8\pi^3 k m^3 e^2}$  (C)  $\frac{n^3 h^3}{8\pi^2 k m^2 e^2}$  (D)  $\frac{n^2 h^2}{8\pi^2 k m^2 e^2}$

Q.10 Which of the following figure correctly represents the electronic transition in the atom of target metal tungsten W for the production of X-rays in the Coolidge tube. (Here jumping direction is shown by the vertical line.)



### [PARAGRAPH TYPE]

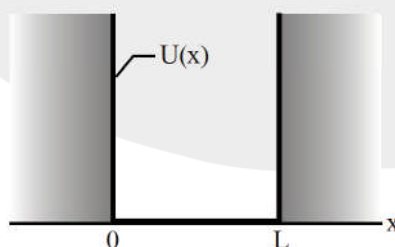
#### Paragraph for question nos. 11 to 13

Figure shows the potential energy of the electron as a function of its position along the x-axis of the idealized trap. When the electron is in the central cylinder, its potential energy  $U$  (in eV) is zero because there the potential  $V$  is zero. If the electron could get outside this region, its potential energy would be positive and of infinite magnitude because there  $V \rightarrow -\infty$ . We call the potential energy pattern of figure an infinitely deep potential energy well or, for short, an infinite potential well.

Just like the standing wave in a length of stretched string, the matter wave describing the confined electron must have nodes at  $x = 0$  and  $x = L$ . By using de-Broglie's hypothesis,

We get the energy  $E$ . We find that  $E$  depends on  $n$  according to

$$E_n = \left( \frac{h^2}{8m_e L^2} \right) n^2, \quad \text{for } n = 1, 2, 3, \dots$$



Perhaps the most direct way to construct a potential energy well in the laboratory is to prepare a sample of a semiconducting material in the form of a powder whose granules are small and of uniform size. Each such granule - each crystallite - acts as a potential well for the electrons trapped within it.

Equation shows that we can increase the energy level values of an electron trapped in an infinite well by reducing the width  $L$  of the well. This would also shift the photon energies that the well can absorb to higher values and thus shift the corresponding wavelengths to shorter values. A given crystallite can absorb photons with an energy above a certain threshold energy  $E_t (= hf_t)$  and thus wavelengths below a corresponding threshold wavelength

$$\lambda_t = \frac{c}{f_t} = \frac{hc}{E_t}$$

Light with any wavelength longer than  $\lambda_t$  is scattered by the crystallite instead of being absorbed. The color we attribute to the crystallite is then determined by the wavelength composition of the scattered light we intercept.



The photograph here shows Semiconductor cadmium selenide, each consisting of a powder of crystallites of uniform size. The lower sample scatters light at the red end of the spectrum. The upper sample differs from the lower sample only in that the upper sample is composed of smaller crystallites.

Thus, the sample now scatters both red and yellow. Because the yellow component happens to be brighter, the sample's color is now dominated by the yellow.

- Q.11 What is the threshold energy  $E_t$  indicated in the paragraph?
- (A)  $\frac{h^2}{8mL^2}$  (B)  $\frac{h^2}{2mL^2}$  (C)  $\frac{h^2}{mL^2}$  (D)  $\frac{3h^2}{8mL^2}$
- Q.12 If we use crystallites of cadmium selenide which are bigger than the lower sample, it could
- (A) Absorb red light and scatter infrared light  
 (B) Absorb yellow light and scatter red light.  
 (C) Absorb blue light and scatter yellow light.  
 (D) Absorb violet light and scatter green light.
- Q.13 Assume that the lower sample has a threshold wavelength of 620 nm. Estimate the order of size of the crystallite.
- (A)  $10^{-10}$  m (B)  $10^{-9}$  m (C)  $10^{-11}$  m (D)  $10^{-8}$  m

### [REASONING TYPE]

- Q.14 **Statement-1:** Cut off wavelength of X-ray coming from a Coolidge tube depends on the target material.  
**Statement-2:** The characteristic X-rays are produced when fast moving electrons remove one of the electrons in the inner shells of the target atoms.
- (1) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.  
 (2) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.  
 (3) Statement-1 is true, statement-2 is false.  
 (4) Statement-1 is false, statement-2 is true.

### [MATRIX TYPE]

- Q.15 If  $Q$  is energy released in the decay,  $m_x$  is atomic mass of parent nucleus,  $m_y$  is atomic mass of daughter nucleus and  $m_e$  is mass of electron, then match the following :

| Column-I                 | Column-II                       |
|--------------------------|---------------------------------|
| (A) $k$ electron capture | (P) $Q = (m_x - m_y)c^2$        |
| (B) $\beta^-$ decay      | (Q) $Q = (m_x - m_y - m_e)c^2$  |
| (C) $\beta^+$ decay      | (R) $Q = (m_x - m_y - 2m_e)c^2$ |
|                          | (S) $Q = (m_x - m_y + 2m_e)c^2$ |

### [SUBJECTIVE TYPE]

- Q.16 A gas containing hydrogen like ions with atomic number  $Z$ , emits photons in transition  $n + 2 \rightarrow n$ , where  $n = Z$ . These photons fall on a metallic plate and eject electrons having minimum de-Broglie wavelength  $\lambda$  of  $5\text{\AA}$ . Find the value of  $Z$  if the work function of metal is  $4.2\text{eV}$ .
- Q.17 There are two bodies A & B with temperatures  $1500\text{ K}$  and  $3600\text{ K}$  respectively. Photons having wavelength equal to the wavelength corresponding to maximum spectral radiance of A & B (say  $\lambda_A$  and  $\lambda_B$  respectively) are incident on a metal sphere of radius  $12\text{ cm}$  having work-function  $1.2\text{ eV}$ . The maximum kinetic energies of the electrons emitted by these photons are in the ratio  $1 : 3$ .
- (a) Find the wavelengths  $\lambda_A$  and  $\lambda_B$ .
- (b) If  $10^{12}$  photons of wavelength  $\lambda_B$  are incident per second and photo electric efficiency is one electron per  $10^6$  photons, find the time after which emission of photo-electrons will stop. In this case assume that  $\lambda_A$  is not present.
- Q.18 A nucleus of  $^{60}\text{Ni}$  in an excited state decays to its ground state by emitting  $1.33\text{ MeV}$  photon. What is the recoil energy and recoil speed of the  $^{60}\text{Ni}$  nucleus ?
- Q.19 A perfectly absorbing surface intercepts a parallel beam of monochromatic light of  $\lambda = 500\text{ nm}$ , incident on it normally. If the power through any cross-section of beam is  $10\text{ W}$ .
- (i) Find the number of photons absorbed per second by the surface.
- (ii) Find the force exerted by light beam on the surface.
- Q.20 A solution contains a mixture of isotopes  $X^A$  of half life  $14\text{ days}$  and  $X^B$  of half life  $25\text{ days}$ . Total activity is  $1\text{ curie}$  at time  $t = 0$ . The activity reduces by  $50\%$  in  $20\text{ days}$ .
- (a) Find the initial activities of  $X^A$  &  $X^B$ .
- (b) Find the ratio of their initial numbers of nuclei.

### ANSWER KEY

|      |                                                                |      |                   |      |                                                  |      |   |      |                                  |                         |   |     |   |
|------|----------------------------------------------------------------|------|-------------------|------|--------------------------------------------------|------|---|------|----------------------------------|-------------------------|---|-----|---|
| Q.1  | C                                                              | Q.2  | B                 | Q.3  | D                                                | Q.4  | B | Q.5  | D                                | Q.6                     | B | Q.7 | A |
| Q.8  | C                                                              | Q.9  | A                 | Q.10 | B                                                | Q.11 | D | Q.12 | A                                | Q.13                    | A |     |   |
| Q.14 | 4                                                              | Q.15 | (A)-P (B)-P (C)-R |      |                                                  |      |   |      |                                  |                         |   |     |   |
| Q.16 | The value of $Z=2$ (neglecting the negative/ fractional value) |      |                   |      |                                                  |      |   |      | Q.17                             | 30940 Å, 1289 Å, 7000 S |   |     |   |
| Q.18 | 15.7 eV, 7 km/s                                                |      |                   | Q.19 | $2.52 \times 10^{19}$ eV, $3.3 \times 10^{-8}$ N |      |   | Q.20 | (a) 0.367 curie (b) 0.633 curie] |                         |   |     |   |

